The Aronhold-Kennedy theorem: A graphical proof

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March 8, 2020

0.1 Relative motion of three planes

With reference to Figure 1, let us denote by:

- Π_i, Π_j and Π_k three planes in coplanar relative motion;
- P_{ij}, P_{ik}, P_{jk} are the instant center of the relative motions $\Pi_i - \Pi_j$, $\Pi_i - \Pi_k$, $\Pi_j - \Pi_k$, respectively;
- ω_{ij}, ω_{ik}, ω_{jk} are the angular velocities of the relative motions $\Pi_i - \Pi_j$, $\Pi_i - \Pi_k$, $\Pi_j - \Pi_k$, respectively;
- M^i and M^j two instantaneously overlapped points of planes Π_i and Π_j, respectively.

Without loss of generality, Π_k is assumed as kinematic frame from which the remaining planar motions are observed.

For an observer on Π_k, the velocities of M_i and M_j can be respectively expressed as:

$$\vec{v}_{M^i} = \vec{\omega}_{ik} \times \overrightarrow{P_{ik}M^i} \tag{1}$$

$$\vec{v}_{M^j} = \vec{\omega}_{jk} \times \overrightarrow{P_{jk}M^j} \tag{2}$$

If we require coincidence of M^i (or M^j) with P_{ij}, then $\vec{v}_{M^i} = \vec{v}_{M^j}$ (no relative motion between M^i and M^j). Therefore, P_{ij} lies necessarily on the line joining P_{ik} and P_{jk} otherwise the two previous velocities will be not parallel (necessary condition to be equal).

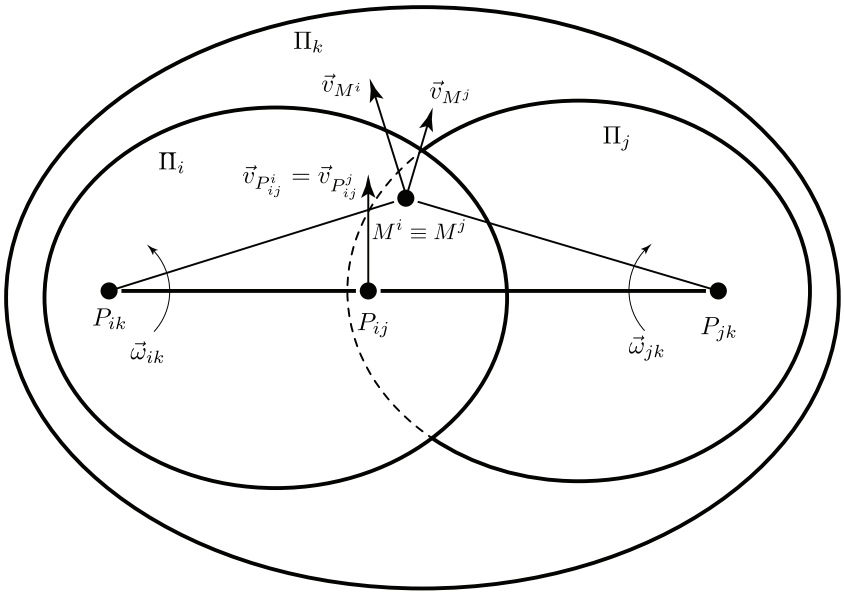


Figure 1: Aronhold-Kennedy theorem

This proves the Aronhold - Kennedy theorem:

Theorem 0.1.1 *The velocity poles of three bodies in relative coplanar motion lie on the same straight line.*

Instant centers play a fundamental role in the understanding of the planar motion kinematics.

This theorem is very important in applications such as the determination of velocity ratios and for the evaluation of mechanical advantage in mechanisms.

In Figure 2 are reported some applications of the Aronhold-Kennedy theorem.

Cases a) and b) of Figure 2 - Velocity ratios in a four-bar linkage:

The velocities of points P_{23} considered on bodies 2 and 3 are, respectively:

$$\vec{v}_{P^1_{23}} = \vec{\omega}_{21} \times \overrightarrow{P_{12}P_{23}}$$
$$\vec{v}_{P^2_{23}} = \vec{\omega}_{31} \times \overrightarrow{P_{13}P_{23}}$$

Since there is not relative motion between these points $\vec{v}_{P^1_{23}} = \vec{v}_{P^2_{23}}$ and the following angular velocity ratio can be established:

$$\frac{\omega_{21}}{\omega_{31}} = \pm\frac{P_{13}P_{23}}{P_{12}P_{23}} \tag{3}$$

with the plus (minus) sign valid when P_{23} is outside (inside) segment $P_{12}P_{13}$, as in case b) (a).

Considering the relative motion of links 2 and 4, the velocities of points P_{24} considered on bodies 2 and 4 are, respectively:

$$\vec{v}_{P^2_{24}} = \vec{\omega}_{21} \times \overrightarrow{P_{12}P_{24}}$$
$$\vec{v}_{P^4_{24}} = \vec{\omega}_{41} \times \overrightarrow{P_{14}P_{24}}$$

Since there is not relative motion between these points $\vec{v}_{P^2_{24}} = \vec{v}_{P^4_{24}}$ and the following angular velocity ratio can be established:

$$\frac{\omega_{21}}{\omega_{41}} = \pm\frac{P_{14}P_{24}}{P_{12}P_{24}} \tag{4}$$

with the plus (minus) sign valid when P_{24} is outside (inside) segment $P_{12}P_{14}$, as in Case a) (b).

Case c) of Figure 2 - Velocity ratio in a slider-crank linkage:

Since the slider undergoes an absolute translational motion (*i.e.* all points of body 4 have the same velocity), the following equality holds

$$\vec{v}_{P^2_{24}} = \vec{v}_s \tag{5}$$

with v_s absolute slider velocity.

The absence of relative motion in P_{24} gives the condition

$$\vec{v}_{P^2_{24}} = \vec{v}_{P^4_{24}} \tag{6}$$

Hence, combining the previous equation with (5), one obtains:

$$\vec{v}_s = \vec{\omega}_{21} \times \overrightarrow{P_{12}P_{24}} \tag{7}$$

Case d) of Figure 2 - Velocity ratio in a rotating slider linkage:

Since there is not relative motion between P^2_{23} and P^3_{23}, then

$$\vec{\omega}_{21} \times \overrightarrow{P_{12}P_{23}} = \vec{\omega}_{31} \times \overrightarrow{P_{13}P_{23}} \tag{8}$$

Case e) of Figure 2 - Velocity ratio in rotating levers:
Since there is not relative motion between P_{23}^2 and P_{23}^3, then

$$\vec{\omega}_{21} \times \overrightarrow{P_{12}P_{23}} = \vec{\omega}_{31} \times \overrightarrow{P_{13}P_{23}} \tag{9}$$

Case f) of Figure 2 - Velocity ratio in cam mechanism:
Since there is not relative motion between P_{23}^2 and P_{23}^3, then

$$\vec{\omega}_{21} \times \overrightarrow{P_{12}P_{23}} = \vec{v}_{P_{23}^3} \tag{10}$$

with $\vec{v}_{P_{23}^3}$ is the translation velocity.

For some linkages with eight or more links, the Aronhold-Kennedy theorem alone does not allow the complete determination of all instant centers. Such linkages are commonly known as *indeterminate linkages*. Instant center which can be found by inspection are referred as a *primary instant center*. The remaining ones are referred to as a *secondary instant center*.

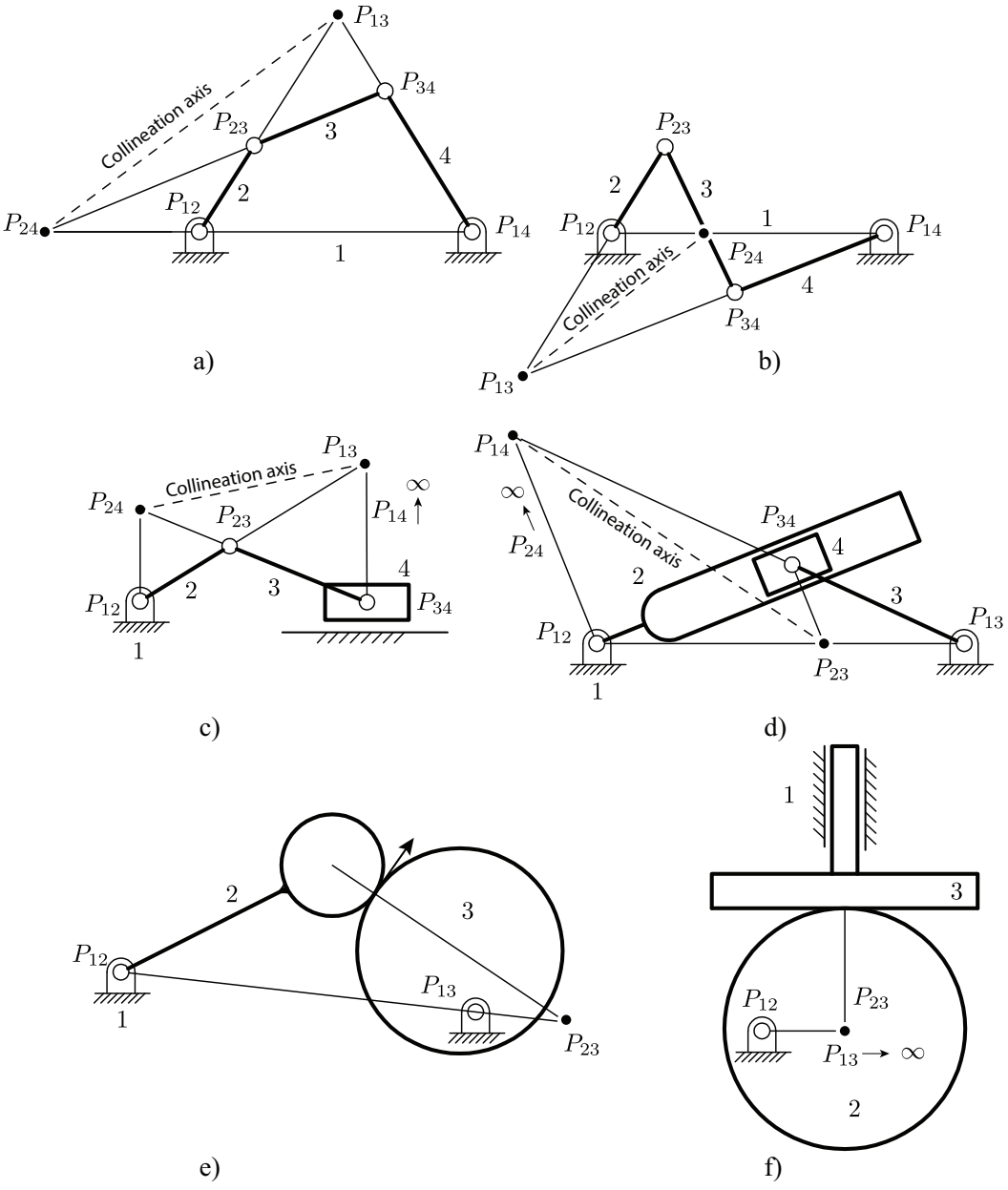


Figure 2: Applications of the Aronhold-Kennedy theorem

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